PythonX Bootcamp

Python Foundations for Data Science

Assignment 01 **Total Marks: 20**

Instructions

- Answer all questions.
- Write **Python code with proper comments**.
- Use **clear variable names** and **good programming practices**.
- Submit your work as a **.py file or Jupyter/Colab Notebook (.ipynb)**.

Question 1: You have joined a **data science research team** working on an AI model for disease prediction. The team wants a small Python script to confirm that the development environment is working properly.Write a Python program that: **(3 Marks)**

1. Prints the message **"Python Environment Ready for Data Science!"**
2. Displays your **Name, ID, University, and Research Area** using formatted output.
3. Add **comments explaining the purpose of the script**.

Question 2: A data scientist is preparing a dataset for a machine learning project. The dataset contains information about: **(3 Marks)**

- Dataset Name
- Number of Samples
- Model Accuracy from a previous experiment

Write a Python program that:

1. Stores this information in **variables**.
2. Prints the information in a **readable format**.
3. Displays the **data type of each variable**.

This task helps verify **data types before processing datasets**.

Question 3: During model testing, different models produce different accuracy scores. The research team wants a quick classification of model performance. **(3 Marks)**

Write a Python program that:

1. Takes **model accuracy as input**.

2. Uses **conditional statements** to classify performance as:
- **Excellent Model** → accuracy ≥ 90
- **Good Model** → 75 ≤ accuracy < 90
- **Poor Model** → accuracy < 75
3. Print the result with a clear message.

Question 4: A machine learning engineer is working with multiple dataset folders. Before processing, they want to review the dataset names. Given the following dataset list: ["train_images", "validation_images", "test_images", "corrupted_dataset", "ecg_dataset"] **(3 Marks)**

Write a Python program that:

1. Stores the dataset names in a **list**.
2. Uses a **for loop** to display each dataset.
3. If the dataset name is "corrupted_dataset", skip processing it using **continue**.

Question 5: A research assistant needs a function to compute the **average accuracy of multiple experiments**. Write a Python function named: calculate_experiment_accuracy() **(4 Marks)**

The function should:

1. Accept **three accuracy scores as parameters**.
2. Calculate the **average accuracy**.
3. Return the result.
4. Display the average accuracy with a proper message.

Example:

Experiment Accuracies: 88, 91, 93
Average Accuracy = 90.67
Question 6: A data science team maintains metadata files describing datasets. **(4 Marks)**

Write a Python program that:

1. Creates a file named **dataset_metadata.txt**
2. Writes the following information:
   - ☐ Dataset Name: PythonX_ID
   - ☐ Type: Time-Series Medical Data
   - ☐ Number of Samples: 109446
3. Reads and prints the content of the file.
4. Use **try–except–finally** to handle possible file errors.